Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2026

Course: B.Tech | *Time: 03 Hours*
Code: 100817 | *Subject: Cloud Computing* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct option / answer the following (Any seven question only):* **[2 x 7 = 14]**

(a) Cloud computing mainly provides
(i) Localized computing (iii) Offline processing
(ii) Shared computing resources (iv) Manual scheduling

(b) Which service model provides virtual machines and storage?
(i) SaaS (iii) IaaS
(ii) PaaS (iv) DaaS

(c) A company requires 20 virtual machines each with 4 GB RAM. Total RAM required is
(i) 40 GB (iii) 80 GB
(ii) 60 GB (iv) 100 GB

(d) Which hypervisor runs directly on hardware?
(i) Type-1 Hypervisor (iii) Container Engine
(ii) Type-2 Hypervisor (iv) Virtual Switch

(e) If 5 tasks require 10 ms each and are executed sequentially, total execution time is
(i) 10 ms (iii) 50 ms
(ii) 20 ms (iv) 100 ms

(f) Spot instances in cloud computing are mainly used for
(i) High-cost reserved workloads (iii) Permanent storage
(ii) Low-cost temporary workloads (iv) Security enhancement

(g) Which cloud model is owned by a single organization?
(i) Public Cloud (iii) Private Cloud
(ii) Community Cloud (iv) Hybrid Cloud

(h) Fog computing mainly reduces
(i) Latency (iii) Storage
(ii) Encryption (iv) Virtualization

(i) The major security issue in multi-tenancy is
(i) Data isolation (iii) CPU speed
(ii) Compilation error (iv) Routing delay

(j) Advance reservation in cloud computing means
(i) Dynamic scheduling (iii) Deleting resources
(ii) Reserving resources before use (iv) VM shutdown

Q.2 (a) Explain cloud computing architecture and deployment models with neat diagrams. **[7]**
(b) Differentiate between SaaS, PaaS, and IaaS with suitable examples. **[7]**

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Q.3	(a)	Explain virtualization in cloud computing and virtualization issues.	[7]
	(b)	Discuss multi-tenancy issues and their challenges.	[7]
Q.4	(a)	Explain the Amazon EC2 and Amazon S3 services and their architectures.	[7]
	(b)	Discuss the implementation of private and hybrid clouds using open-source tools.	[7]
Q.5	(a)	Discuss the architecture and working of hypervisors in cloud computing.	[7]
	(b)	Discuss different load balancing algorithms used in cloud computing.	[7]
Q.6	(a)	Explain resource optimization algorithms in cloud computing.	[7]
	(b)	Explain Quality of Service parameters in cloud computing.	[7]
Q.7	(a)	Explain task migration and VM migration techniques.	[7]
	(b)	Discuss cloud security threats and vulnerabilities.	[7]
Q.8	(a)	Discuss application-level, data-level, and VM-level security in cloud computing.	[7]
	(b)	Explain Green Cloud and IoT with suitable examples.	[7]

Q.9 Write short note on *any two* of the following: [7 x 2 = 14]

(a) Mobile Cloud Computing
(b) Infrastructure Security
(c) SLA Management
(d) Fog Computing

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